

An Infinite-Product Counterexample for Rokhlin Dimension

Candidate negative answers to Questions 5.2 and 5.3 of arXiv:1407.1277

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Status. Candidate full counterexamples, likely valid, requiring expert review. The construction is a constant-order version of the infinite product suggested in the source.

Abstract

Let $G = \prod_{j \geq 1} \mathbb{Z}/2\mathbb{Z}$ act on the infinite torus $X = \prod_{j \geq 1} S^1$ by coordinatewise signs. The action is free and the induced action on $C(X)$ has the X -Rokhlin property. We prove that its Rokhlin dimension is nevertheless infinite. For the quotient $H_i = (\mathbb{Z}/2)^i$, the associated principal bundle squares the first i circle coordinates. Its mod-2 kernel cup length is at least i , hence its Schwarz genus and equivariant index are at least i . A published inverse-limit formula then makes the Rokhlin dimension the supremum of these indices. Since $C(X)$ is commutative, the commuting-tower dimension is infinite as well. This gives simultaneous negative answers to Questions 5.2 and 5.3 as stated.

1 The source questions

Gardella asks whether finite Rokhlin dimension with commuting towers is equivalent to the X -Rokhlin property for arbitrary compact groups, and whether, in the commutative case, finite Rokhlin dimension is equivalent to freeness of the induced action on the maximal ideal space [1, Questions 5.2–5.3]. The source immediately suggests an infinite product of cyclic rotations but leaves its Rokhlin dimension unchecked.

tion 4.19), then conditions (a), (b), (c), (f) and (g) are equivalent by Corollary 4.20

- (3) If A is a Kirchberg algebra and $G = \mathbb{Z}_2$ (and possibly also if G is any finite group), then (e) and (f) are equivalent by Theorem 2.3 in [2].

5. OUTLOOK AND OPEN PROBLEMS

In this last section, we give some indication of possible directions for future work, and raise some natural questions related to our findings.

Although some of our results, particularly in Section 4, assume that the acting group is a Lie group, this is probably not needed everywhere. Our first suggested problem is then:

Problem 5.1. Extend some of the results in this paper to actions of not necessarily finite-dimensional compact groups.

We point out that the assumption that G be a Lie group in Corollary 4.3 in [13] is necessary, since it relies on Atiyah-Segal completion Theorem (see [1], and see Theorem 1.1.1 in [28]), for which one needs the representation ring $R(G)$ to be finitely generated. We suspect that Corollary 4.6 is not true in general for arbitrary compact groups (it probably already fails for actions on compact spaces), but it may be the case that all compact group actions with finite Rokhlin dimension with commuting towers have *locally* discrete K -theory.

Somewhat related, we ask:

Question 5.2. Is finite Rokhlin dimension with commuting towers equivalent to the X -Rokhlin property for *arbitrary* compact groups?

Maybe one should start with the commutative case:

Question 5.3. Is finite Rokhlin dimension for actions on commutative C^* -algebras, equivalent to freeness of the induced action on the maximal ideal space for *arbitrary* compact groups?

For compact Lie groups, the answer is yes in both cases; see Theorem 4.5 and Theorem 4.2. It was crucial in the proofs of said theorems that free actions of compact Lie groups have local cross-sections. We suspect that the answer to Question 5.2 and Question 5.3 is ‘no’, and it is possible that the action of $G = \prod_{n \in \mathbb{N}} \mathbb{Z}_n$ on $X = \prod_{n \in \mathbb{N}} S^1$ by coordinate-wise rotation is a counterexample (to both questions). Such action has the X -Rokhlin property and is free, but the quotient map $X \rightarrow X/G$ is known not to have local cross-sections, since X is not locally homeomorphic to $X/G \times G$. We have not checked, however, whether this action has finite Rokhlin dimension.

Figure 1. Questions 5.2 and 5.3 and the proposed infinite-product route in the source, PDF p. 28.

Transcription. Question 5.2 asks whether finite Rokhlin dimension with commuting towers is equivalent to the X -Rokhlin property for arbitrary compact groups. Question 5.3 asks whether finite Rokhlin dimension for actions on commutative C^* -algebras is equivalent to freeness on the maximal ideal space for arbitrary compact groups.

2 Proof intuition

Finite Rokhlin dimension can be tested on the finite Lie quotients of a compact metrizable group. The product group $G = (\mathbb{Z}/2)^\mathbb{N}$ has quotients $H_i = (\mathbb{Z}/2)^i$. For the coordinatewise action on the infinite torus, the H_i -bundle seen at level i is the product of i nontrivial double covers. Each double cover kills one degree-one mod-2 class, but the product of all i classes survives on the base. Sectional-category cup length therefore forces at least $i + 1$ local sections, in the convention that the genus is at least i . These lower bounds diverge with i , so the inverse-limit formula forces infinite Rokhlin dimension. Freeness and the X -Rokhlin property, by contrast, are immediate from the coordinate action and the identity map on $C(X)$.

3 The counterexample

Write $\mathbb{Z}_2 = \{1, -1\}$. Set

$$G = \prod_{j=1}^{\infty} \mathbb{Z}_2, \quad X = \prod_{j=1}^{\infty} \mathbb{T},$$

and define

$$(g_j)_{j \geq 1} \cdot (z_j)_{j \geq 1} = (g_j z_j)_{j \geq 1}.$$

Both G and X are compact metrizable, and the action is continuous.

Theorem 1. *Let $\alpha : G \rightarrow \text{Aut}(C(X))$ be the induced action. Then:*

- (i) *the action of G on X is free;*
- (ii) *α has the X -Rokhlin property;*
- (iii) $\dim_{\text{Rok}}(\alpha) = \dim_{\text{Rok}}^c(\alpha) = \infty$.

Consequently Questions 5.2 and 5.3 of [1] both have negative answers.

We first record the elementary cohomological estimate used below. We use the convention of [2]: $\text{gen}(p) \leq m$ means that the base is covered by $m + 1$ open sets on each of which the bundle admits a section.

Lemma 1 (Kernel cup-length bound). *Let $p : E \rightarrow B$ be a principal bundle. If there are positive-degree classes $v_1, \dots, v_k \in \ker(p^* : H^*(B; R) \rightarrow H^*(E; R))$ with $v_1 \smile \dots \smile v_k \neq 0$, then $\text{gen}(p) \geq k$.*

Proof. Suppose $\text{gen}(p) \leq k - 1$, and choose a sectional cover $B = U_1 \cup \dots \cup U_k$. If $s_r : U_r \rightarrow E$ is a section, then

$$v_r|_{U_r} = s_r^* p^* v_r = 0.$$

Thus v_r lifts to $H^*(B, U_r; R)$. The relative cup product of these lifts lies in $H^*(B, U_1 \cup \dots \cup U_k; R) = H^*(B, B; R) = 0$, so $v_1 \smile \dots \smile v_k = 0$, a contradiction. $\square$

Proof of Theorem 1. If $g \cdot x = x$, then $g_j x_j = x_j$ for every j . Since $x_j \in \mathbb{T}$, it follows that $g_j = 1$ for every j , so the action is free.

The X -Rokhlin property asks for an approximate equivariant central unital homomorphism from $C(X)$, equivalently an equivariant unital homomorphism into the central sequence algebra [3, Definition 1.5]. Here the identity homomorphism $C(X) \rightarrow C(X)$, followed by the constant-sequence inclusion, already has all these properties. Hence α is X -Rokhlin.

For $i \geq 1$, let

$$N_i = \{g \in G : g_1 = \cdots = g_i = 1\} \cong \prod_{j>i} \mathbb{Z}_2, \quad H_i = G/N_i \cong (\mathbb{Z}_2)^i.$$

Then $N_1 \supset N_2 \supset \cdots$, $\bigcap_i N_i = \{e\}$, and $G = \varprojlim H_i$. Put $Y_i = X/N_i$ and $B = X/G$. Using $\mathbb{T}/\{\pm 1\} \cong \mathbb{T}$, with quotient coordinate $z \mapsto z^2$, identify both Y_i and B with $\mathbb{T}^{\mathbb{N}}$. Under these identifications the quotient of the free H_i -action is

$$q_i : Y_i \longrightarrow B, \quad q_i(z_1, z_2, \dots) = (z_1^2, \dots, z_i^2, z_{i+1}, z_{i+2}, \dots).$$

Let η be the generator of $H^1(\mathbb{T}; \mathbb{F}_2)$, and let $u_j = \pi_j^* \eta \in H^1(B; \mathbb{F}_2)$, where π_j is the j -th coordinate projection. For $1 \leq j \leq i$, naturality and the degree-two circle map give

$$q_i^* u_j = 2 \pi_j^* \eta = 0 \quad \text{in } H^1(Y_i; \mathbb{F}_2).$$

On the other hand,

$$u_1 \smile \cdots \smile u_i \neq 0.$$

Indeed, restrict along the inclusion $\mathbb{T}^i \hookrightarrow \mathbb{T}^{\mathbb{N}}$ obtained by setting all later coordinates equal to 1; the restriction is the nonzero top Künneth class of $\mathbb{T}^i$. Lemma 1 therefore gives

$$\text{gen}(q_i) \geq i.$$

For compact Hausdorff group actions on compact Hausdorff spaces, Schwarz genus equals the equivariant index [2, Proposition 4.4]. Moreover, the inverse-limit formula of [2, Corollary 5.15] gives

$$\dim_{\text{Rok}}(\alpha) = \sup_i \text{ind}_{H_i}(X \times_G H_i) = \sup_i \text{ind}_{H_i}(X/N_i) = \sup_i \text{gen}(q_i) \geq \sup_i i = \infty.$$

Thus $\dim_{\text{Rok}}(\alpha) = \infty$. Finally, all ranges in the central sequence algebra of the commutative algebra $C(X)$ commute, so ordinary and commuting-tower Rokhlin dimensions agree. Hence $\dim_{\text{Rok}}^c(\alpha) = \infty$ as well.

The free action with infinite Rokhlin dimension disproves the equivalence in Question 5.3. The same action is X -Rokhlin but has infinite commuting-tower Rokhlin dimension, disproving the equivalence in Question 5.2. $\square$

4 Verification, scope, and novelty

The proof audit checked compact metrizability, continuity and freeness, the exact X -Rokhlin definition, the finite-quotient bundle coordinates, the relative-cohomology cup-length argument, equality of genus and index, and every hypothesis of the inverse-limit formula. No computation is used. The supporting formula occurs as Theorem 5.14 and Corollary 5.15 of [2]; that paper does not appear to compute this product example.

The source suggested $G = \prod_n \mathbb{Z}_n$ acting on $\prod_n S^1$. We use the cleaner constant-order group $\prod_j \mathbb{Z}_2$, for which arbitrarily long mod-2 cup products give a uniform lower-bound mechanism. This settles the two yes/no questions but does not address the source's other open problems.

A bounded search on 11 August 2026 covered all four run indexes, exact question wording, the arXiv id and title, the published source, the author's thesis, X -Rokhlin and arbitrary-compact-group phrases, coordinatewise rotations, infinite Rokhlin dimension, Schwarz genus, arXiv:1801.00767, and later citations returned by those searches. It found the original suggestion and the later general formula but no explicit calculation of this example or announced resolution of Questions 5.2–5.3. Novelty is therefore provisional.

Human-review recommendation. Check the identification of q_i , the genus convention in Lemma 1, and the application of Corollary 5.15 of [2]. Also confirm that Question 5.2 uses the X -Rokhlin property with an arbitrary compact free G -space, as in [3]; the underlying X then supplies the required witness.

References

- [1] E. Gardella, *Rokhlin dimension for compact group actions*, Indiana Univ. Math. J. **66** (2017), no. 2, 659–703; arXiv:1407.1277.
- [2] E. Gardella, P. M. Hajac, M. Tobolski, and J. Wu, *The local-triviality dimension of actions of compact quantum groups*, arXiv:1801.00767v3 (2019). See Proposition 4.4, Theorem 5.14, and Corollary 5.15.
- [3] I. Hirshberg and N. C. Phillips, *Rokhlin dimension: obstructions and permanence properties*, Doc. Math. **20** (2015), 199–236, especially Definition 1.5; arXiv:1410.6581.